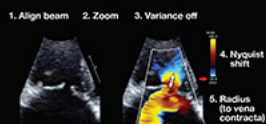




Tricuspid Regurgitation

A. Color Flow Doppler (2D and 3D)

- Proximal Flow Convergence**
1. Align direction of flow with insonation beam
 2. Zoomed view
 3. Variance off
 4. Change baseline of Nyquist limit in the direction of the jet and adjust to obtain hemispheric flow convergence (typically ≥ 20 cm/s)
 5. Measure the radius (white arrow in image) from the point of color aliasing to the vena contracta



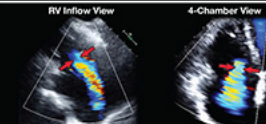
Advantage:

- Rapid qualitative assessment

Disadvantages:

- Multiple jets
- Non-hemispheric shape

- Vena Contracta**
1. Zoomed view
 2. Apical 4-ch view
 3. RV inflow view



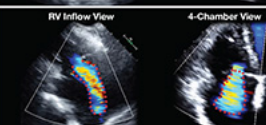
Advantages:

- Surrogate for regurgitant orifice size
- Independent of flow rate and driving pressure for a fixed orifice
- Less dependent on technical factors
- Good at identifying severe TR (>0.7 cm)

Disadvantages:

- Underestimates severity with multiple jets
- Imaging of convergence zone for measurement

- Jet Area**
1. 4-ch RV inflow or subcostal views



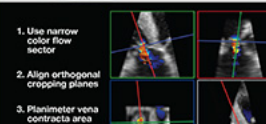
Advantage:

- Easy to measure

Disadvantages:

- Dependent on the driving pressure and jet direction
- Direction and shape of jet may overestimate (central entrainment) or underestimate (eccentric, wall-impinging) jet area

- 3D Vena Contracta**
1. Color flow sector should be narrow
 2. Align orthogonal cropping planes along the axis of the jet
 3. Choose a mid-systolic cycle and planimeter vena contracta area



Advantage:

- Multiple jets of differing directions may be measured

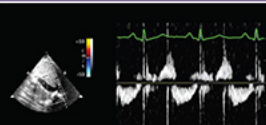
Disadvantages:

- Dynamic jets may be over- or underestimated
- Time consuming
- Limited spatial resolution will lead to overestimation

Note: Non-coaxial jets or aliased flow may appear "laminar" but still represent regurgitant flow

B. Pulsed Wave Doppler

- Hepatic Vein Flow Reversal**
1. Align insonation beam with the flow in the hepatic vein



Advantages:

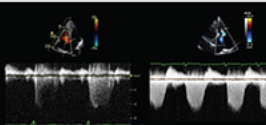
- Simple supportive sign of severe TR
- Can be obtained with both TTE and TEE

Disadvantages:

- Depends on compliance of the right atrium
- May not be reliable in patients with atrial fibrillation, paced rhythm with retrograde atrial conduction

C. Continuous Wave Doppler

- Density of Regurgitant Jet**
1. Align insonation beam with the flow



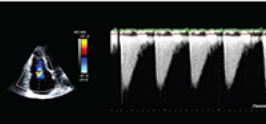
Advantages:

- Simple
- Density is proportional to the number of red-blood cells reflecting the signal
- Faint or incomplete jet is compatible with mild TR

Disadvantages:

- Qualitative
- Perfectly central jets may appear denser than eccentric jets of higher severity
- Overlap between moderate and severe TR

- Jet Contour**
1. Align insonation beam with the flow



Advantages:

- Simple
- Specific sign of pressure equalization in low velocity, early peaking dense TR jet

Disadvantages:

- Qualitative
- Affected by changes that modify RV and RA pressures

D. Quantitative Doppler: EROA, Regurgitation Volume

- PISA:**
1. Align direction of flow with insonation beam
 2. Zoomed view
 3. Variance off
 4. Change baseline of Nyquist limit in the direction of the jet and adjust to obtain hemispheric flow convergence (typically ≥ 20 cm/s)
 5. Measure the radius (white arrow from the point of color aliasing to the vena contracta)
 6. CW Doppler of regurgitant jet for peak velocity and VTI



Advantage:

- Quantitative assessment of lesion severity (EROA) and volume overload (RVol)

Disadvantages:

- Not valid for multiple jets, less accurate in eccentric jets
- Limited experience and evidence
- Typically lower RV pressures than LV lead to greater contour flattening and underestimation in proportion to the ratio of the aliasing velocity to the peak TR velocity

Grading the Severity of Chronic TR by Echocardiography¹

Parameters	Mild	Moderate	Severe
Structural			
TV morphology	Normal or mildly abnormal leaflets	Moderately abnormal leaflets	Severe valve lesions (e.g., flail leaflet, severe retraction, large perforation)
RV and RA size	Usually normal	Normal or mild dilation	Usually dilated ²
Inferior vena cava diameter	Normal <2cm	Normal or mildly dilated 2.1-2.5cm	Dilated >2.5cm
Quantitative Doppler			
Color flow jet area ³	Small, narrow, central	Moderate central	Large central jet or eccentric wall-impinging jet of variable size
Flow convergence zone	Not visible, transient or small	Intermediate in size and duration	Large throughout systole
CWD jet	Faint/partial/parabolic	Dense, parabolic or triangular	Dense, often triangular
Semi-quantitative			
Color flow jet area (cm ²) ⁴	Not defined	Not defined	>10
VCI (cm ²) ⁵	<0.3	0.3-0.69	≥0.7
PISA radius (cm) ⁶	≤0.5	0.6-0.9	≥0.9
Hepatic vein flow ⁷	Systolic dominance	Systolic blunting	Systolic flow reversal
Tricuspid inflow ⁸	A-wave dominant	Variable	E-wave >1.0m/sec
Quantitative			
EROA (cm ²)	<0.20	0.20-0.39 ⁹	≥0.40
RVol (mL/beat)	<30	30-44 ⁹	≥45

RA, Right Atrium

1. Boxed signs are considered specific for their TR grade.

2. RV and RA size can be within the "normal" range in patients with acute severe TR.

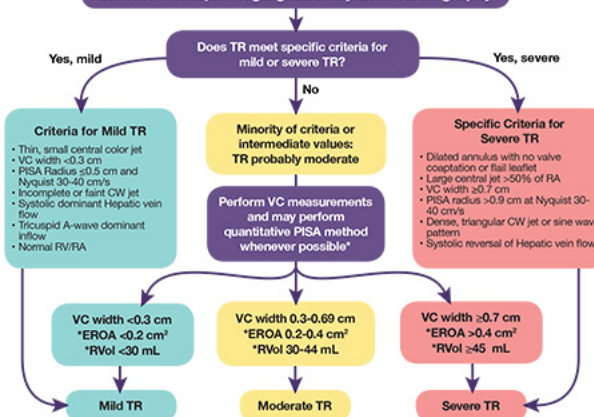
3. With Nyquist limit >50-70 cm/sec.

4. With baseline Nyquist limit shift of 20 cm/sec.

5. Signs are nonspecific and are influenced by many other factors (RV diastolic function, atrial fibrillation, RA pressure).

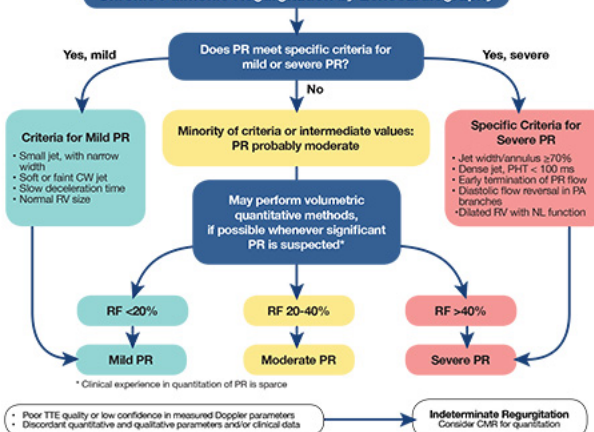
6. There are little data to support further separation of these values.

Chronic Tricuspid Regurgitation by Echocardiography



* Clinical experience in quantitation of TR is sparse

Chronic Pulmonic Regurgitation by Echocardiography



* Clinical experience in quantitation of PR is sparse

• Poor TTE quality or low confidence in measured Doppler parameters
• Discrepant quantitative and qualitative parameters and/or clinical data

Indeterminate Regurgitation
Consider CWR for quantitation